Mancala Adversarial Search — Heuristic Evaluation Report

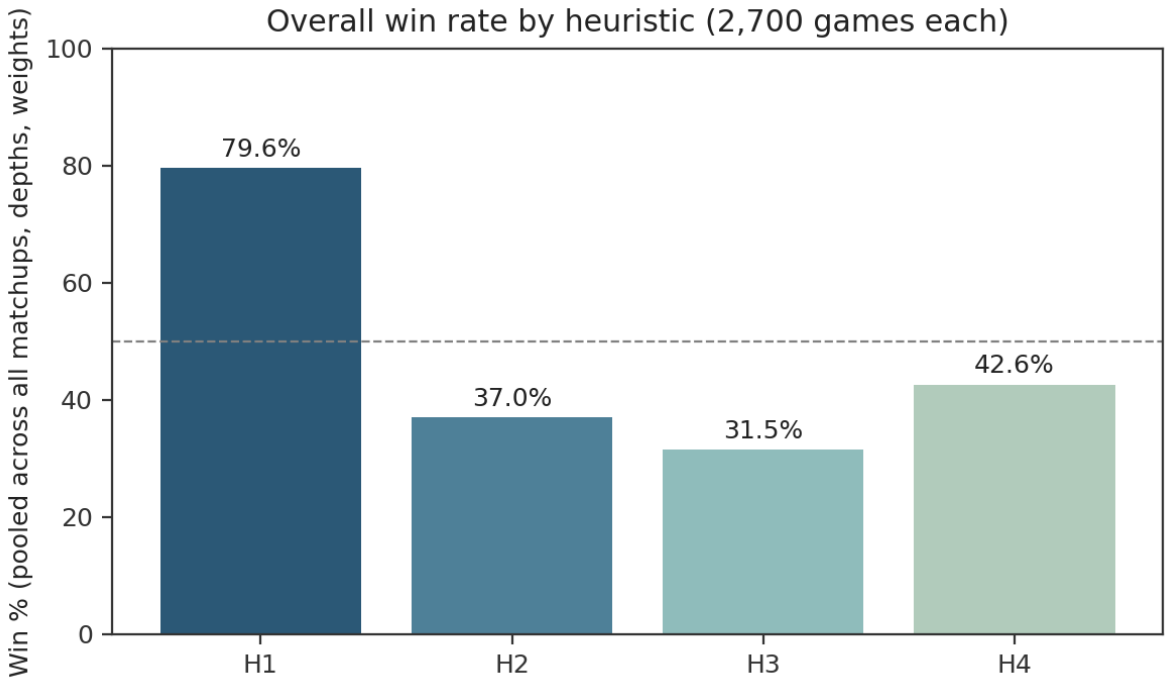
Minimax with alpha-beta pruning. 4 heuristics (H1-H4, cumulative), 6 pairwise matchups, 3 search depths (3/5/7), 3 weight sets (default, equal, store-heavy). 100 games per configuration, first-move alternated between the two heuristics each game. 5,400 games total.

Heuristic definitions

ID	Formula
H1	store_diff
H2	w1*store_diff + w2*side_diff
H3	w1*store_diff + w2*side_diff + w3*extra_move_opportunities
H4	w1*store_diff + w2*side_diff + w3*extra_move_opportunities + w4*capture_opportunity_value

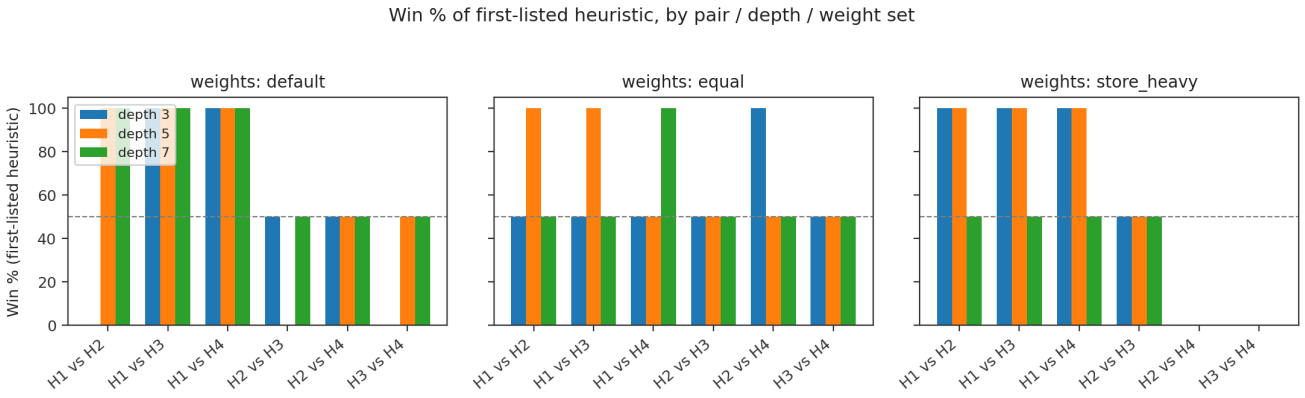
weight_label	w1	w2	w3	w4
default	1.0	2.0	3.0	4.0
equal	1.0	1.0	1.0	1.0
store_heavy	3.0	1.0	1.0	1.0

Overall win rate by heuristic (pooled, 2,700 games each)



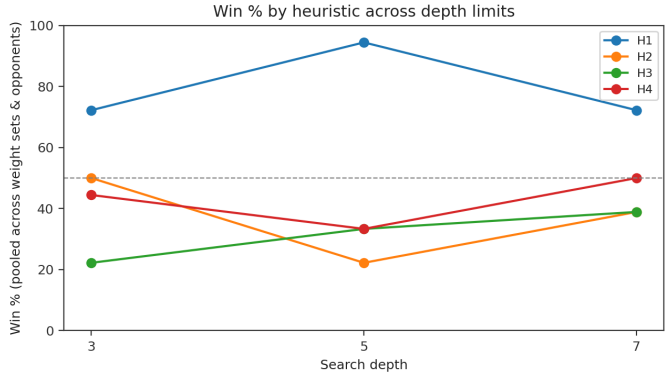
heuristic	games	wins	draws	win_pct	draw_pct
H1	2700	2150	50	79.63%	1.85%
H2	2700	1000	100	37.04%	3.70%
H3	2700	850	150	31.48%	5.56%
H4	2700	1150	200	42.59%	7.41%

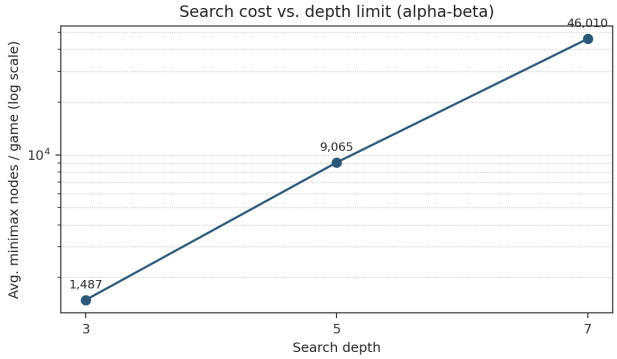
Win % by pair, depth, and weight set



Bars show win % of the first-listed heuristic in each pair (e.g. H1 in "H1 vs H2"). Dashed line = 50%.

Win % by heuristic across depth limits





depth	avg nodes/game
3	1,487
5	9,065
7	46,010

Left: win % per heuristic pooled across weight sets/opponents, by depth. Right: mean alpha-beta node count per game, by depth (log-scale plot + raw means).

Full results — all 54 configurations

pair	depth	weights	games	A_win	B_win	draw	A_win%	B_win%	draw%	avg_nodes
H1 vs H2	3	default	100	0	100	0	0%	100%	0%	1,880
H1 vs H3	3	default	100	100	0	0	100%	0%	0%	1,711
H1 vs H4	3	default	100	100	0	0	100%	0%	0%	1,914
H2 vs H3	3	default	100	50	50	0	50%	50%	0%	987
H2 vs H4	3	default	100	50	50	0	50%	50%	0%	2,072
H3 vs H4	3	default	100	0	100	0	0%	100%	0%	1,648
H1 vs H2	3	equal	100	50	50	0	50%	50%	0%	1,396
H1 vs H3	3	equal	100	50	50	0	50%	50%	0%	1,402
H1 vs H4	3	equal	100	50	50	0	50%	50%	0%	1,675
H2 vs H3	3	equal	100	50	0	50	50%	0%	50%	882
H2 vs H4	3	equal	100	100	0	0	100%	0%	0%	1,090

pair	depth	weights	games	A_win	B_win	draw	A_win%	B_win%	draw%	avg_nodes
H3 vs H4	3	equal	100	50	0	50	50%	0%	50%	1,323
H1 vs H2	3	store_heavy	100	100	0	0	100%	0%	0%	1,328
H1 vs H3	3	store_heavy	100	100	0	0	100%	0%	0%	1,328
H1 vs H4	3	store_heavy	100	100	0	0	100%	0%	0%	1,477
H2 vs H3	3	store_heavy	100	50	50	0	50%	50%	0%	1,376
H2 vs H4	3	store_heavy	100	0	100	0	0%	100%	0%	1,594
H3 vs H4	3	store_heavy	100	0	100	0	0%	100%	0%	1,680
H1 vs H2	5	default	100	100	0	0	100%	0%	0%	11,158
H1 vs H3	5	default	100	100	0	0	100%	0%	0%	9,146
H1 vs H4	5	default	100	100	0	0	100%	0%	0%	9,778
H2 vs H3	5	default	100	0	100	0	0%	100%	0%	8,258
H2 vs H4	5	default	100	50	0	50	50%	0%	50%	9,264
H3 vs H4	5	default	100	50	0	50	50%	0%	50%	6,873
H1 vs H2	5	equal	100	100	0	0	100%	0%	0%	9,632
H1 vs H3	5	equal	100	100	0	0	100%	0%	0%	11,258
H1 vs H4	5	equal	100	50	0	50	50%	0%	50%	7,722
H2 vs H3	5	equal	100	50	50	0	50%	50%	0%	8,152
H2 vs H4	5	equal	100	50	50	0	50%	50%	0%	7,679
H3 vs H4	5	equal	100	50	50	0	50%	50%	0%	7,870
H1 vs H2	5	store_heavy	100	100	0	0	100%	0%	0%	9,570
H1 vs H3	5	store_heavy	100	100	0	0	100%	0%	0%	9,622
H1 vs H4	5	store_heavy	100	100	0	0	100%	0%	0%	9,800
H2 vs H3	5	store_heavy	100	50	50	0	50%	50%	0%	12,180
H2 vs H4	5	store_heavy	100	0	100	0	0%	100%	0%	8,532
H3 vs H4	5	store_heavy	100	0	100	0	0%	100%	0%	6,680
H1 vs H2	7	default	100	100	0	0	100%	0%	0%	50,116
H1 vs H3	7	default	100	100	0	0	100%	0%	0%	45,338
H1 vs H4	7	default	100	100	0	0	100%	0%	0%	64,120
H2 vs H3	7	default	100	50	50	0	50%	50%	0%	42,166
H2 vs H4	7	default	100	50	50	0	50%	50%	0%	33,434
H3 vs H4	7	default	100	50	50	0	50%	50%	0%	26,042
H1 vs H2	7	equal	100	50	50	0	50%	50%	0%	49,074
H1 vs H3	7	equal	100	50	50	0	50%	50%	0%	44,482
H1 vs H4	7	equal	100	100	0	0	100%	0%	0%	59,050
H2 vs H3	7	equal	100	50	50	0	50%	50%	0%	26,488
H2 vs H4	7	equal	100	50	50	0	50%	50%	0%	62,080
H3 vs H4	7	equal	100	50	50	0	50%	50%	0%	40,113
H1 vs H2	7	store_heavy	100	50	50	0	50%	50%	0%	46,367
H1 vs H3	7	store_heavy	100	50	50	0	50%	50%	0%	47,252
H1 vs H4	7	store_heavy	100	50	50	0	50%	50%	0%	53,997

pair	depth	weights	games	A_win	B_win	draw	A_win%	B_win%	draw%	avg_nodes
H2 vs H3	7	store_heavy	100	50	50	0	50%	50%	0%	37,422
H2 vs H4	7	store_heavy	100	0	100	0	0%	100%	0%	49,452
H3 vs H4	7	store_heavy	100	0	100	0	0%	100%	0%	51,193

A = first-listed heuristic in pair, B = second-listed. Win % computed over 100 games with starting player alternated each game.